

Naïve bayes

Friday, 27 June 2025

8:45 PM



Estimating
Probabiliti...

⇒ The Data follows some non-standard (standard) distribution.

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↳ We can approximate this distribution by some standard distribution & apply optimal Bayes on this distribution to get good classification.

→ MLE: $\theta = \underset{\theta}{\operatorname{argmax}} P(D|\theta)$ (θ is the parameter which sets the standard distribution to a distribution close to our standard model)

→ MAP: $\theta = \underset{\theta}{\operatorname{argmax}} P(\theta|D)$ (Given that we observe the data, what is the most likely θ).
↳ for this we need to define the prior distribution $P(\theta)$.

⇒ After selecting a distribution, we can then find the prediction.

∴ Prediction $P(y|x=x)$. → the distribution is fixed.

The general form is:-

$$P(y|x=x) = \int_0 P(y|\theta) P(\theta|D) d\theta$$

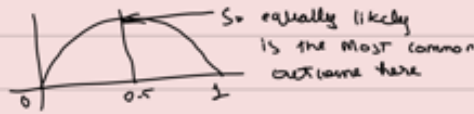
↳ the prediction averages out all possible models.

⇒ When the prior is a constant distribution, MAP & MLE behaves the same

✱

✱ After large number of sample, regardless of the prior itself learn MLE & MAP converges to same point.

- ⇒ The prior is set up on a beta distribution usually.
 ↳ ex:- I assume, that I've seen at least 1 heads & 1 tails, so in β distribution $\alpha=1, \beta=1$



- ✱ If prior is aggressive/sharper plot, MAP needs much more points to converge.
 ⇒ demo in Lecture 8, from 10:00 (Awesome Demo!)
- ✱ MAP estimator is good when you've little data, but if you have a dead prior, it's not good at all.
 ↳ If the prior belief is good then, MAP will converge much-much faster

⇒ The problem:-

$$P(Y|X=x)$$

⇒ If x is 1-D or has a small dimension this would work, but if it has millions of dimensions then the probability that

$$P(X=x) = P(X_1=x_1, X_2=x_2, \dots, X_{\infty}=x_{\infty})$$

 which is very small/rarely the case.

o Naïves Bayes:-

⇒ The labels on the other hand are limited to few 100 dimensions.

Using Bayes rule, they flip $P(Y|X=x)$ as $P(X|Y=y_i)$.

$$\therefore P(Y=y|X=x) = \frac{P(X|Y=y) \cdot P(Y=y)}{P(X=x)}$$

$P(X=x|Y=y)$ is still difficult because

$P(X_1=x_1, X_2=x_2, \dots, X_{m+1}=x_{m+1}|Y=y)$ is still very small.

✱ But, $P(X_1=x_1|Y=y), P(X_2=x_2|Y=y), \dots, P(X_n=x_n|Y=y)$ are very significant

$\therefore$ Naive Bayes assumes that each value of $P(X_i = x_i | Y = y)$ is independent for each i .

$$\therefore P(x = x_i | y = y) = \prod_{\alpha=1}^d P(X_\alpha = x_\alpha | y = y).$$

$$\therefore h(\vec{x}) = \operatorname{argmax}_y P(y | \vec{x}) = \operatorname{argmax}_y \frac{P(x | y) \cdot P(y)}{P(x)}.$$

(x) $\rightarrow$ doesn't contain y so removed.

$$\therefore \operatorname{argmax}_y P(x | y) \cdot P(y)$$

$$\therefore \operatorname{argmax}_y P(y) \cdot \prod_{\alpha} P(x_\alpha | y).$$

$$= \operatorname{argmax}_y \log P(y) + \sum_{\alpha} \log P(x_\alpha | y)$$

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* If the assumption that one feature (dimension) is independent of others than my assumption holds true, & then Naive Bayes becomes Bayes optimal classifier, which is the most ideal classifier to ever exist.

$\Rightarrow$ Other assumption is about the probability distribution.

1) We can have $X_\alpha = \langle F_1, \dots, F_k \rangle$, i.e. each feature can take value 1 to k .

$$P(X_\alpha = j | y = c) = [\theta_{jc}]_\alpha, \sum_{j=1}^k [\theta_{jc}]_\alpha = 1$$

$\therefore$ a feature can take any one of the k values
summing over all the possible values of j
 $j \in \{1, \dots, k\}$

• Multinomial distribution:-

$\Rightarrow x_\alpha \in \{0, 1, \dots, m\}$
 $\hookrightarrow$ frequency of α words.

'a' : 5
 'ant' : 0
 'all' : 0 } total words $m = 5$

$\therefore m = \sum_{\alpha=1}^d x_\alpha \rightarrow$ frequency of the word α .

$P(\vec{x} | m, y=c) = \frac{m!}{x_1! \dots x_d!} \prod_{\alpha=1}^d \theta_{\alpha c}^{x_\alpha}$
 $\hookrightarrow$ given the frequency
 $\theta_{\alpha c}$ } Probability of drawing the word α .

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$\Rightarrow$ In Naive Bayes instead of fully estimating probability density, which is very hard, we make a very simple model assumption, we assume that these different dimensions are independent, so in some sense we're projecting data to those axes & model a combined distribution by simply multiplying all independent distribution.

• $P(x|y) = P(y) \Rightarrow$ we need to find this

$\begin{pmatrix} \text{'o'} \\ \text{'ant'} \\ \text{'Hi'} \\ \text{'X'} \\ \text{'pumpkin'} \\ \text{'animal'} \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 0 \\ 5 \\ 0 \\ 1 \end{pmatrix} \in \mathbb{R}^d$

$$\Rightarrow P(\mathcal{X} | m, y = c) = \left(\prod_{\alpha=1}^m (\theta_{\alpha c})^{x_{i\alpha}} \right).$$

$$\hat{\theta}_{\alpha c} \leftarrow \frac{\sum_{i=1}^n I(y_i = c) x_{i\alpha}}{\sum_{i=1}^n I(y_i = c) \left(\sum_{\beta=1}^d [x_i]_{\beta} \right)}.$$

If you want to smooth

$$\hat{\theta}_{\alpha c} = \frac{\sum_{i=1}^n I(y_i = c) x_{i\alpha} + 1}{\sum_{i=1}^n I(y_i = c) \left(\sum_{\beta=1}^d [x_i]_{\beta} \right) + d}$$

Q1 Why divide by d where d is the words in the dictionary?

→ Because when we smooth we're saying that every word exist at least once across (all the emails i.e. if n emails then word i must exist at least once in atleast one of the emails).

Q2 In practice $P(y = \text{spam} | x)$ & $P(y \neq \text{spam} | x)$ both have same denominator.

So we divide them $\frac{P(y = \text{spam} | x)}{P(y \neq \text{spam} | x)}$

if > 1 then spam

< 1 then not spam.

Q11 what if my x 's are continuous?

Q12 $x \in \mathbb{R}$ $\therefore$ here we use gaussian distributions.

$$P(x_{\alpha}|y) = N(\mu_{\alpha c}, \sigma_{\alpha c}^2).$$

$\hookrightarrow$ different mean & variance for each feature & each class.

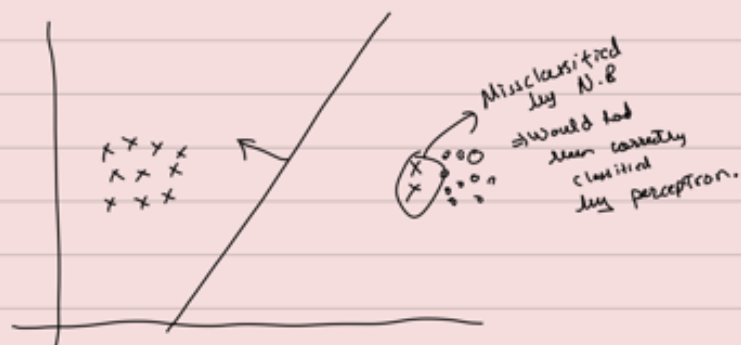
Let $n_c = \sum_{i=1}^n I(y_i = c)$ $\therefore$ is number of points in the class.

$$\mu_{\alpha c} = \frac{1}{n_c} \sum_{i=1}^n I(y_i = c) x_{i\alpha c} \quad \text{Mean}$$

$$\sigma_{\alpha c}^2 = \frac{1}{n_c} \sum_{i=1}^n I(y_i = c) (x_{i\alpha c} - \mu_{\alpha c})^2$$

Q13 Perceptron always gives us a separating hyperplane if there exists one, Naïve Bayes doesn't always give . . . 'even' . . . '!'.

$\hookrightarrow$ ex:- If the NB is modelled on a gaussian, then if the real data is not gaussian then it's a trouble.



* Naive Bayes as a Linear Classifier:-

$$\Rightarrow P(y|x) \propto P(x|y) \cdot P(y).$$

$P(y=+1|x) > P(y=-1|x) \Rightarrow$ If this is true which decision boundary exists.

$$P(x|y=+1) \cdot P(y=+1) > P(x|y=-1) \cdot P(y=-1)$$

$$\Rightarrow P(+1) \left(\frac{m!}{x_1! \dots x_n!} \right) \prod_{\alpha=1}^d \theta_{\alpha(+1)}^{x_{\alpha}} > P(-1) \left(\frac{m!}{x_1! \dots x_n!} \right) \prod_{\alpha=1}^d \theta_{\alpha(-1)}^{x_{\alpha}}$$

$$\therefore \log P(+1) + \sum_{\alpha=1}^d x_{\alpha} \log \theta_{\alpha(+1)} > \log P(-1) + \sum_{\alpha=1}^d x_{\alpha} \log \theta_{\alpha(-1)}$$

$$\therefore \underbrace{\log P(+1) - \log P(-1)}_b + \sum_{\alpha=1}^d x_{\alpha} \underbrace{\left(\log \theta_{\alpha(+1)} - \log \theta_{\alpha(-1)} \right)}_{w_{\alpha}} > 0$$

$$\Rightarrow w^T x + b > 0$$

i.e. A linear classifier.

★ In most common cases NB gives linear classifier, otherwise it's not a linear classifier, when $P(x|c)$ are from exponential families. The NB corresponds to a linear classifier in feature space.

o When to use NB over Perceptron?

→ When both classes are not linearly separable, perceptron will give bad solutions.

→ Perceptron is good if they are linearly separable.

⇒ If you do the same thing for Gaussian Naive Bayes:-

$$p(y|\vec{x}) = \frac{1}{1+e^{-w_2}} \quad y \in \{+1, -1\}.$$